

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

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COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems. (BL3, BL6)*

Activity Tool : Analytical Problem Solving (High)

Assessment Tool Weightage: 35%

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QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Apply relational algebra operations (σ , π , $\cup$, $-$, $\times$, $\bowtie$) on the given Employee and Department relations to retrieve required records.	BL3 – Apply	5
2	Write SQL DDL statements to create STUDENT(SID, Name, DOB, DeptID) and DEPARTMENT(DeptID, DeptName) tables with all appropriate constraints.	BL3 – Apply	5
3	Formulate SQL queries using GROUP BY, HAVING, and aggregate functions (COUNT, SUM, AVG, MAX, MIN) on a given Sales dataset.	BL3 – Apply	5
4	Write a correlated sub-query to find employees earning more than the average salary of their own department.	BL3 – Apply	5
5	Apply all types of JOIN operations (INNER, LEFT OUTER, RIGHT OUTER, FULL OUTER, CROSS) on Employee and Project tables.	BL3 – Apply	5
6	Create a view 'DeptSalaryView' that displays department name, total employees, and average salary. Write a query on this view.	BL6 – Create	5
7	Construct a relational algebra expression for the following: Find names of students who are enrolled in all courses offered.	BL6 – Create	5

8	Write SQL DML statements (INSERT, UPDATE, DELETE) with integrity constraint enforcement for a given scenario.	BL3 – Apply	5
9	Apply tuple relational calculus to express a query: Find all employees who work in the 'IT' department and earn more than 50000.	BL3 – Apply	5
10	Design a relational schema for an Airline Reservation System and write 5 SQL queries demonstrating joins, sub-queries, and views.	BL6 – Create	5

Code of Conduct:

*I JAISURIYA D D(192421167) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q1. Relational Algebra Operations

Assume:

EMPLOYEE(EID, Name, Salary, DeptID)

DEPARTMENT(DeptID, DeptName)

a) Employees with salary > 50000

$\sigma_{\text{Salary} > 50000}(\text{EMPLOYEE})$

b) Display only employee names

$\pi_{\text{Name}}(\text{EMPLOYEE})$

c) Employee details with department names

$\text{EMPLOYEE} \bowtie \text{DEPARTMENT}$

d) Cartesian Product

$\text{EMPLOYEE} \times \text{DEPARTMENT}$

e) Employees not in Department 101

$\text{EMPLOYEE} - \sigma_{\text{DeptID}=101}(\text{EMPLOYEE})$

Q2. SQL DDL Statements

SQL

CREATE TABLE Department(

DeptID INT PRIMARY KEY,

DeptName VARCHAR(30) NOT NULL

);

CREATE TABLE Student(

SID INT PRIMARY KEY,

Name VARCHAR(50) NOT NULL,

DOB DATE,
DeptID INT,
FOREIGN KEY(DeptID)
REFERENCES Department(DeptID)
);

Q3. GROUP BY, HAVING and Aggregate Functions

Assume:

Sales(SaleID, Product, Amount)

Count Products

SQL

SELECT Product,COUNT(*)

FROM Sales

GROUP BY Product;

Total Sales

SQL

SELECT Product,SUM(Amount)

FROM Sales

GROUP BY Product;

Average Sales

SQL

SELECT Product,AVG(Amount)

FROM Sales

GROUP BY Product;

Maximum Sale

SQL

SELECT MAX(Amount)

FROM Sales;

HAVING Clause

SQL

SELECT Product,SUM(Amount)

FROM Sales

GROUP BY Product

HAVING SUM(Amount)>10000;

Q4. Correlated Subquery

SQL

SELECT Name,Salary

FROM Employee E

WHERE Salary>(

SELECT AVG(Salary)

FROM Employee

WHERE DeptID=E.DeptID

);

Q5. JOIN Operations

INNER JOIN

SQL

SELECT *

FROM Employee E

INNER JOIN Project P

ON E.EID=P.EID;

LEFT JOIN

SQL

SELECT *

FROM Employee E

LEFT JOIN Project P

ON E.EID=P.EID;

RIGHT JOIN

SQL

SELECT *

FROM Employee E

RIGHT JOIN Project P

ON E.EID=P.EID;

FULL OUTER JOIN

SQL

SELECT *

FROM Employee E

FULL OUTER JOIN Project P

ON E.EID=P.EID;

CROSS JOIN

SQL

SELECT *

FROM Employee

CROSS JOIN Project;

Q6. View

Create View

SQL

```
CREATE VIEW DeptSalaryView AS
```

```
SELECT D.DeptName,
```

```
COUNT(E.EID) AS TotalEmployees,
```

```
AVG(E.Salary) AS AverageSalary
```

```
FROM Department D
```

```
JOIN Employee E
```

```
ON D.DeptID=E.DeptID
```

```
GROUP BY D.DeptName;
```

Query View

SQL

```
SELECT *
```

```
FROM DeptSalaryView;
```

Q7. Relational Algebra

Find students enrolled in all courses.

$$\pi \text{ Name}(\text{Student} \bowtie (\text{Enrollment} \div \text{Course}))$$

(If your faculty has not taught the division operator, ask whether they expect an equivalent expression, because the assessment only states the requirement and not the notation to use.)

CO2- 1 Analytical problem Solvi...

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Q8. SQL DML

INSERT

SQL

INSERT INTO Student

VALUES(101,'Rahul','2005-05-20',1);

UPDATE

SQL

UPDATE Student

SET Name='Arun'

WHERE SID=101;

DELETE

SQL

DELETE FROM Student

WHERE SID=101;

Q9. Tuple Relational Calculus

Employees working in IT department with salary above 50000.

$\{ t \mid \text{Employee}(t)$

$\wedge t.\text{Dept} = \text{'IT'}$

$\wedge t.\text{Salary} > 50000 \}$

Q10. Airline Reservation System

Tables

Passenger

PassengerID

PassengerName

Phone

Flight

FlightID

FlightName

Source

Destination

Booking

BookingID

PassengerID

FlightID

SeatNo

JourneyDate

SQL Queries

1. Join

SQL

SELECT PassengerName,FlightName

FROM Passenger

JOIN Booking

ON Passenger.PassengerID=Booking.PassengerID

JOIN Flight

ON Booking.FlightID=Flight.FlightID;

2. Subquery

SQL

SELECT PassengerName

FROM Passenger

WHERE PassengerID IN

(

SELECT PassengerID

FROM Booking

);

3. Count Bookings

SQL

SELECT COUNT(*)

FROM Booking;

4. Create View

SQL

CREATE VIEW FlightDetails AS

SELECT FlightName,Source,Destination

FROM Flight;

5. Query View

SQL

SELECT *

FROM FlightDetails;